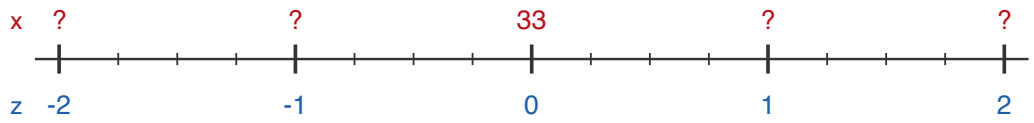
# Z-Values for Commute Times

The commute times of students to campus have mean $\mu = 33$ minutes and standard deviation $\sigma = 4$ minutes.

The z-value of a commute time $x$ is

$$z = \frac{x - \mu}{\sigma}$$

a) Fill in the x-values for each z-value on the number line.

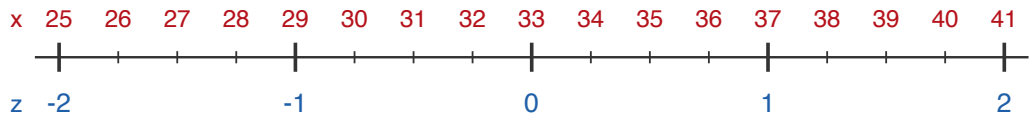



For each commute time below:

- **Visually:** mark $x$ on the number line and read off its z-value.
- **Calculate:** find the z-value with the formula.

Check that your two answers match.

**b) Ana's commute is $x = 39$ minutes**




Visually: $z =$ _____

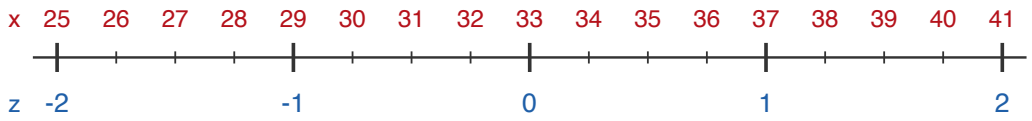
Calculate: $z = \frac{39 - 33}{4} =$ _____

**c) Ben's commute is $x = 31$ minutes**

Visually: $z =$ _____

Calculate: $z =$ _____

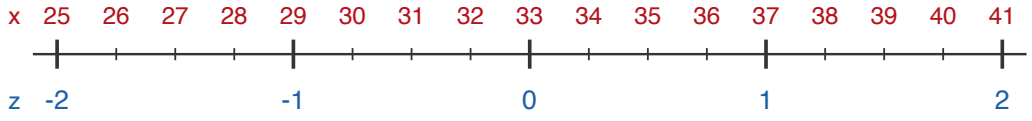
**d) Carla's commute is $x = 26$ minutes**




Visually: $z =$ _____

Calculate: $z =$ _____

**e) Dev's commute is $x = 36$ minutes**




Visually: $z =$ _____

Calculate: $z =$ _____